

Risk-Managed Time-Series Momentum in Crypto Majors: Crash-State De-Risking and Drawdown Control

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Abstract

We study a fixed-lookback ($L = 30$ day) long/cash time-series-momentum (TSMOM) sleeve on liquid crypto-majors (BTC, ETH, BNB, XRP, ADA, SOL, DOGE), overlaid with a market-drawdown-state de-risking rule. The construction follows Moskowitz, Ooi and Pedersen (2012): hold each asset only while its trailing 30-day cumulative log-return is positive, otherwise rotate that allocation to cash; then scale the whole book's gross exposure down ($\times 0.5$) when the aggregate crypto market is more than 15% off its trailing peak. The headline finding is a drawdown-control result, not an alpha result. Across an offline 96-configuration grid, 100% of configurations produced a smaller maximum drawdown than equal-weight buy-and-hold; in an out-of-sample (OOS) walk-forward test with the lookback fixed a priori at $L = 30$, the sleeve delivered net-of-25bps Sharpe 1.41 (Harvey-Liu-Zhu $t = 3.21$) versus buy-and-hold Sharpe 0.70 ($t = 1.60$), with maximum drawdown cut roughly in half (-44.6% vs -84.4%). Adding the drawdown-state de-risk overlay ($\times 0.5$ beyond 15% off-peak) further reduced maximum drawdown to about -28% and lifted Calmar from roughly 1.56 to 1.98 while holding Sharpe near 1.53 ($t = 3.69$). The effect survives a frozen long-history (survivorship-hardened) universe and a 50bps round-trip cost assumption. We are explicit that this is *risk-managed β* — a documented trend premium with crash protection — not market-neutral alpha; that vol-targeting / Kelly sizing on top of this sleeve was tested and SETTLED NULL (it de-levers the payoff engine); that the $L = 30$ optimum is a recorded, accepted sharp-parameter fragility; that absolute returns are survivorship-inflated; and that the forward Harvey $t > 3$ promotion bar has NOT yet been cleared (the sleeve is paper-only and forward-accruing).

1. Introduction

A small, concentrated crypto book faces a structural hazard that dominates its compounding outcome: catastrophic drawdown. Equal-weight buy-and-hold of crypto majors over our sample suffers a maximum drawdown near -84% (Sharpe ≈ 0.78 over the full sample, ≈ 0.70 OOS). An 84% peak-to-trough loss requires a 525% subsequent gain merely to recover; for an account that must survive to keep compounding, halving that drawdown is more consequential than chasing incremental return.

This paper does not claim to discover alpha in crypto. A prior internal returns audit (2026-06-01) and the cross-sectional sweeps that preceded this work established that standalone price-momentum in crypto, after beta-neutralization and deflated-Sharpe-ratio (DSR) correction for multiple testing, is NOT a confirmed market-neutral edge: the best beta-neutral cross-sectional configuration reached only Sharpe 0.81 ($t = 1.91$, DSR 0.656), positive in only 3 of 7

years, i.e. bull-year directional leakage through an imperfect beta hedge. Order-flow / microstructure fusion (an OFI-style proxy) was likewise null (best fused Sharpe 0.37, $t = 0.88$, DSR 0.349). Both were settled as honest nulls.

What remains, after those nulls, is a single robust, documented, OOS-validated effect: *risk management* via time-series momentum. Time-series momentum (Moskowitz, Ooi and Pedersen, 2012) is directional-with-risk-management — it rides uptrends and steps aside in downtrends — and its best-documented property is crash protection: trend-followers tend to be out of an asset before the worst of a sustained decline. Crypto is the natural arena: it is the venue with the deepest drawdowns (so the crash-protection property has the most to protect), it is capacity-irrelevant for a small book, and trade frequency (rebalance every 3 days) is unconstrained by pattern-day-trader rules because crypto trades 24/7.

Our contribution is therefore narrow and honest: a frozen, pre-registered, OOS-validated *risk-managed β* sleeve for crypto majors, plus a market-drawdown-state de-risk overlay that materially improves Calmar and drawdown on top of the base sleeve. We report the supporting nulls (sizing, lookback-ensemble, whipsaw filter) with equal prominence, because the design of this sleeve is in large part the residue of what did NOT work.

2. Related Work

Time-series momentum. Moskowitz, Ooi and Pedersen (2012) document time-series momentum across 58 instruments and four asset classes, showing a persistent trend premium and, critically, that trend strategies exhibit positive returns during equity-market extremes (the "crisis alpha" / crash-protection property). Our $L = 30$ long/cash construction is a single-asset-class, long-biased specialization of their framework.

Trend-following as a payoff profile. Rattray, Granger, Harvey and Van Hemert (2020) characterize fixed-weight rebalancing and trend strategies in terms of their option-like payoff: a rebalanced fixed-weight book is structurally short a straddle on relative performance, whereas trend-following is long convexity / long a straddle-like profile, which is precisely why it pays in sustained moves and bleeds in choppy ranges. This framing explains both why our long/cash trend sleeve protects against crashes and why naive sizing overlays (Section 4.3) fail — they truncate exactly the convex payoff that trend-following exists to capture.

Volatility targeting and sizing. Harvey, Hoyle, Korgaonkar, Rattray, Sargaison and Van Hemert (2018, "The Impact of Volatility Targeting") show that volatility targeting improves risk-adjusted returns for assets with a strong leverage effect (where high volatility coincides with low/negative returns), but adds little or can hurt for strategies — such as trend-following — whose own return is already concentrated in high-volatility periods. This is directly relevant: in crypto, the high-volatility periods ARE the high-return trend periods, so vol-targeting de-levers

the payoff engine exactly when it pays. Our sizing-overlay null (Section 4.3) is consistent with their result.

Multiple-testing discipline. Harvey, Liu and Zhu (2016) argue that the conventional $t > 2$ hurdle is far too lenient given the number of strategies tested across the literature, and propose a substantially higher bar (we adopt $t > 3.0$ as the promotion threshold). Bailey and Lopez de Prado (2014) introduce the Deflated Sharpe Ratio (DSR), which discounts an observed Sharpe for the number of independent configurations tried; we report DSR alongside every swept result so that selection bias is visible.

3. Data and Methodology

3.1 Universe and data

The deployable sleeve restricts the trend universe to liquid crypto majors — BTC-USD, ETH-USD, BNB-USD, XRP-USD, ADA-USD, SOL-USD, DOGE-USD — where prior capacity analysis localized the trend edge (large-cap TSMOM Sharpe ≈ 1.15 , $t = 3.38$, passing the Harvey bar; small-alt TSMOM Sharpe ≈ 0.62 , $t = 1.82$, FAILING, due to noise, cost, and survivorship-lottery effects). Offline validation used a broader ≈ 45 -coin universe with point-in-time entry (a coin is included only after it has at least one lookback window of history), and a survivorship-hardened robustness cut on a frozen long-history universe of 41 coins. Daily simple returns over a roughly six-year window were used for the crypto experiments; the multi-asset generality test used 15 ETFs over 15 years.

3.2 Signal construction

The sleeve computes, each rebalance, the trailing-30-day cumulative log-return per asset (lagged to exclude the current day, eliminating look-ahead). Assets with positive trailing trend are held long; the remaining assets' capital is rotated to cash. Capital is budgeted equally across the universe, gross exposure is capped at 1.0 (unlevered), and a long/cash (no-short) construction is used. The primary configuration uses no per-asset volatility scaling (best Sharpe); a conservative variant applies a 0.50 annualized vol target with a $3\times$ per-asset size cap (the drawdown-

minimizing variant). Frozen parameters: lookback = 30 days, rebalance = 3 days, vol window = 30 days. These were pre-registered (2026-06-02) and are not to be tuned without a fresh forward test.

3.3 Drawdown-state de-risk overlay

On top of the per-asset trend sleeve, a single book-level multiplier scales the entire gross exposure. The validated rule de-risks to $\times 0.5$ of full exposure when the aggregate crypto market index is more than 15% off its trailing peak (a "drawdown state"). This is the documented home for the platform's separately-developed market-stress signals: the overlay accepts an external multiplier in $[0, 1]$, so a crash-probability signal (e.g. one minus the crash probability) can be passed in place of, or combined with (most-conservative-wins), the plain drawdown rule. The overlay is a RISK-MANAGEMENT input, not a return-prediction input.

3.4 Validation protocol

We report four nested tests: (P3) a full 96-configuration sweep with DSR deflation; (P4) a walk-forward OOS test contrasting a priori fixed $L = 30$ against in-sample lookback selection; (P5) a survivorship- and cost-robustness cut on the frozen long-history universe at 25 and 50bps; and (P8/P9/P10) the drawdown-state overlay, its module-integration reproduction, and an overlay-threshold robustness sweep. Significance is judged by the Harvey-Liu-Zhu $t > 3.0$ bar on net-of-cost per-period

returns, plus the $DSR > 0.95$ bar where a parameter sweep is involved. Costs are modeled at 10bps (majors) up to 50bps (robustness stress).

4. Results

4.1 Drawdown control is the robust core finding

The single most robust result is the drawdown reduction. In the full 96-configuration sweep (P3), 100% of configurations produced a smaller maximum drawdown than equal-weight buy-and-hold. Buy-and-hold EW crypto returned roughly +63% annualized at Sharpe 0.78 with a -84% maximum drawdown (Calmar 0.75). The best single TSMOM configuration ($L = 30$, $R = 3$, long/cash, 10bps) returned roughly +71% annualized at Sharpe 1.50 ($t = 3.58$), with maximum drawdown -43% (Calmar 1.66). Twelve configurations cleared the $t > 3$ bar — and all twelve were at $L = 30$, i.e. the passing region is consistent, not scattered.

The DSR for the in-sample-best configuration was 0.787, which FAILS the strict 0.95 bar (deflated for 96 trials). We are explicit that this means the *excess-return* claim from the swept maximum is suggestive ($\approx 79\%$ confidence) but not strictly significant; the *drawdown-reduction* claim is the part that is robust to the sweep.

4.2 Out-of-sample: fixed $L = 30$ beats parameter selection

The walk-forward OOS test (P4; train 365d / test 90d roll, net 25bps) confirms the result without selection bias, because the lookback is fixed a priori:

STRATEGY	ANN. RETURN	SHARPE	T-STAT	MAX DD	CALMAR
Fixed $L = 30$ (OOS)	+68.7%	1.41	3.21	-44.6%	1.54
Walk-forward (in-sample L select)	+55.5%	1.10	2.50	-58.2%	0.95
Buy-and-hold (OOS)	+56.5%	0.70	1.60	-84.4%	0.67

Fixed $L = 30$ — zero optimization, the standard one-month lookback — is the honest winner: $t = 3.21$ OOS (passing Harvey), Sharpe and Calmar roughly $2\times$ buy-and-hold, drawdown halved, net of 25bps. Notably, in-

sample lookback selection *underperforms* the fixed choice: the selector picked $L = 30$ in 17 of 21 windows, and the 4 deviations to longer lookbacks

hurt. The operational lesson is to fix the lookback, not optimize it — which also sidesteps the DSR multiple-testing penalty entirely.

4.3 The drawdown-state overlay improves Calmar and drawdown

Stacking the market-drawdown-state de-risk overlay on the base sleeve (P8) is where the largest improvement in tail behavior appears:

CONFIGURATION	SHARPE	T-STAT	MAX DD	CALMAR
TSMOM base	1.46	3.52	-44.6%	1.56
+ drawdown-state de-risk $\times 0.5$ (>15% off peak)	1.53	3.69	-27.6%	1.98
+ drawdown-state de-risk $\times 0.0$	1.29	3.11	-22.1%	1.79
+ vol-percentile de-risk $\times 0.0$	1.53	3.70	-43.6%	1.61

The partial ($\times 0.5$) drawdown-state de-risk lifts Calmar from 1.56 to 1.98 and cuts maximum drawdown from about -45% to about -28%, holding Sharpe near 1.53. Driving the same actual deployable module through the walk-forward (P9) reproduces this — base Sharpe 1.32 / maxDD -52% / Calmar 1.28, and +overlay Sharpe 1.46 ($t = 3.53$), maxDD -29%, Calmar 1.83 — confirming that the production code reproduces the research (live matches backtest). The overlay-threshold sweep (P10, DD-threshold {10-30%} $\times$ de-risk {0-0.75}) found 18 of 20 cells (90%) beat the base on both Calmar and maximum drawdown (median Calmar 1.92 vs base 1.56); only the two "fully-to-cash at a moderate drawdown" cells failed, via whipsaw. The overlay is therefore robust and not cherry-picked; partial de-risk ($\times 0.25$ -0.5) is the sweet spot.

4.4 Survivorship and cost robustness

The survivorship- and cost-hardened cut (P5; fixed $L = 30$ long/cash, full-sample, frozen long-history 41-coin universe, 50bps) holds the core claims: the hardest cell gives $t = 3.11$ with maximum drawdown -48% vs buy-and-hold -84%, Sharpe ≈ 1.3 vs ≈ 0.75 . The drawdown is HALVED in every cell (-45% to -48% vs -84%) — the survivorship-insensitive, rock-solid claim — and Sharpe stays in the 1.3-1.5 range at $t = 3.1$ -3.5 across full-size cells. Importantly, the Calmar edge is thinner than P4 implied (TSMOM 1.26-1.56 vs buy-and-hold 1.21-1.28 full-sample), because full-sample buy-and-hold Calmar is itself

flattered by survivorship-inflated 2020-21 moonshots. The honest headline is therefore drawdown-halving plus Sharpe ($t > 3$), NOT a $2\times$ Calmar gap.

4.5 Multi-asset generality

The crash-protection property generalizes as a drawdown-control overlay but not as a Sharpe-beating alpha (P6; 15 ETFs, 15 years). 100% of multi-asset configurations had a smaller maximum drawdown than equal-weight (best: maxDD -7% vs EW -24% / 60/40 -20%, $t = 3.58$), but the strategy did NOT beat passive 60/40 on Sharpe (0.97 vs 1.08) — it won only on Calmar (0.63 vs 0.48) via the smaller drawdown. The coherent cross-venue picture is that TSMOM's value is crash protection that scales with crash size: large in crypto (drawdown halved, Sharpe ≈ 1.4 vs ≈ 0.7), modest in efficient multi-asset markets (Calmar/drawdown win, Sharpe neutral over a bull-dominated sample). Risk-managed beta everywhere, alpha nowhere.

5. Robustness and Limitations

We state the limitations bluntly; this fund's culture is falsification-rigor, and the negatives are part of the result.

This is risk-managed beta, not market-neutral alpha. The sleeve is long-biased and still loses in sustained bears (offline 2022 about -41%, 2026 about -30%) — roughly half of buy-and-hold, but a loss nonetheless. It is a better way to *hold* crypto, not a market-neutral source of return. Live production corroborates the

broader thesis: the flagship ensemble's headline return decomposed to roughly 98% market beta, with only about 0.42 percentage points of alpha, and the lower-beta model books returned less, not more.

Forward Harvey $t > 3$ is PENDING (paper-only). The promotion rule is hard: no live capital until net-of-cost Harvey $t > 3$ AND the drawdown-reduction claim clears on a FORWARD-PROSPECTIVE sample under the frozen parameters. The provisional quorum (minimum N) is 40 rebalances (≈ 120 trading days); the robust quorum is 120 rebalances. The module computes weights and writes a forward record only; it does NOT place orders. A forward logger has been accruing since 2026-06-02 (rebalance cadence every 3 days); the first live signal was correctly defensive (12 long, 86% cash, market $\approx 53\%$ off peak). Until the forward quorum is met and the bar cleared, every number in this paper is offline / walk-forward, not forward-proven.

The $L = 30$ optimum is a recorded sharp/accepted fragility. All twelve $t > 3$ configurations clustered at $L = 30$; deviations to longer lookbacks degraded the result, and base Calmar drops materially at $L = 20 / L = 45$ in stress extensions. We chose $L = 30$ a priori (the standard one-month lookback) precisely to avoid optimizing into this sharp region, but we record it as an accepted fragility, not a smoothed-over robustness.

Crypto survivorship inflates absolute returns. Delisted alts are absent from the offline universe, so absolute return levels are an upper bound. The drawdown-reduction and the Sharpe-versus-buy-hold improvement are far less sensitive (the same universe sits on both sides of the comparison), which is why we lead with those rather than with absolute return.

Sizing overlays SETTLED NULL — a documented negative. Volatility-targeting and Kelly sizing on top of this sleeve were tested and settled as a DOUBLE NULL. Arm A (cash-dial / Moreira-Muir-style vol targeting) cut annualized return and Calmar across every variant, with negative paired t-stats; Arm B (levered fractional-Kelly) bought higher return at $-51\% / -61\%$ maximum drawdowns (into the knockout zone) with a higher probability of a $>50\%$ drawdown. The mechanism is exactly the Harvey et al. (2018) result: in crypto the high-volatility periods are the

high-return trend periods, so vol-targeting de-levers the payoff engine when it pays. The long/cash sleeve already embeds its own sizing; do not add a sizing layer, and do not acquire leverage for it.

Lookback-ensemble and whipsaw filters SETTLED NULL — further documented negatives. A pre-named equal blend of three lookbacks (20 / 30 / 45) was disconfirmed: in a single-asset-class long/cash book the speeds are collinear (pairwise correlation ≈ 0.93 - 0.95), so the blend bought roughly zero variance reduction while averaging the CAGR down and inheriting the slower speeds' deeper drawdowns (maxDD -36.7% vs -31.0% ; Calmar 1.13 vs 1.78). Multi-speed blending pays with breadth across many MARKETS, not across speeds within one class. A per-asset whipsaw / hysteresis deadband around the trend-sign threshold was also disconfirmed: it halved turnover but dropped gross-of-cost return by roughly $-8.5\%/yr$ from lagging re-entry after V-shaped recoveries, hurting Sharpe (paired Newey-West t about -2.69) with zero timing skill versus a size-matched random-timing control. The stateless immediate-re-entry sign rule IS the engine in cleanly-trending crypto majors.

Cost and threshold caveats. Net results are reported at 10-50bps; the overlay thresholds (15% drawdown, $\times 0.5$) are sensible but lightly chosen on full-sample data, and a forward test is what confirms them. Synthesized fills and absolute return magnitudes should be treated as indicative; the relative (vs buy-and-hold, same-universe) comparisons are the robust ones.

6. Discussion

The honest reading of this work is that crypto did not yield market-neutral alpha to us — cross-sectional momentum, order-flow fusion, and higher-moment signals were all null or DSR-deflated — and that the one durable, documented, OOS-validated edge is risk management. A fixed $L = 30$ long/cash trend sleeve, plus a market-drawdown-state de-risk overlay, roughly halves the maximum drawdown of a crypto-majors book (from about -84% to about -28%), lifts Sharpe from roughly 0.7 to roughly 1.5 ($t > 3$, OOS, net of

cost), and raises Calmar to roughly 2.0, robustly across configurations, costs, and a survivorship-hardened universe.

For a small account, that drawdown reduction is the whole game: the difference between an 84% wipeout (requiring a 525% recovery) and a 28% drawdown (requiring a 39% recovery) is the difference between a knockout and surviving to compound through the cycle. The strategy buys survivorship of *the account*, not of the universe.

The construction is deliberately humble about its mechanism. It does not forecast returns; it manages exposure. This is also the role we assign to the platform's separately-developed market-stress machinery — it plugs into the overlay as a de-risk multiplier (its honest, validated function), not as a return predictor. Whether a richer stress signal de-risks *earlier* than a plain drawdown rule when a real drawdown hits is itself a forward question, and is being shadow-A/B-tested rather than asserted.

The path to any live-capital claim runs entirely through the forward test. The offline and walk-forward evidence is sufficient to justify *running* the forward test under frozen parameters; it is not sufficient to deploy capital. We will report the forward verdict — confirmation or null — when the quorum is met, with the same multiple-testing discipline applied here.

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